Generative model: categorical states, observations, actions, and hierarchy I

The 9 studies — including the contaminated-sentinel robustness sweep (Study 4) — run on one shared sentinel world (Studies 5–7 use its moving and hierarchical variants): a discrete sentinel partially-observable Markov decision process (POMDP) using the categorical world structure illustrated by Friston et al. (2024), Figures 1 and 4. We adopt the discrete-state active-inference formulation that the community has standardized around — the categorical $A/B/C/D_0$ generative model of da Costa et al. (2020), the same object the `pymdp` (Heins et al. 2022) and RxInfer (Bagaev et al. 2023) toolboxes operate on — and reimplement it in pure NumPy in `pomdp.py` so the colony, its sensors, and its dynamics are exactly the ones the analysis executes. A colony of sentinels watches a single hidden creature whose location is the shared latent factor they federate beliefs about.

The structural map in fig. 1 follows the same generative-model vocabulary while making the implementation boundary visible: Panel A shows what one agent actually sees — one noisy categorical report over the 9 possible locations — while Panel B identifies the $A/B/C/D_0$ factors that turn that report into a local posterior. Temporal depth then describes state, observation, posterior, and control order; hierarchical depth describes conditioned priors. It is a formal schematic, not an assertion that every displayed dependency is simultaneously estimated in every study.

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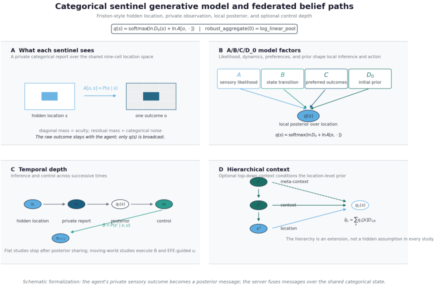

Figure 1: Formal categorical generative-model schema. Source relation: source-inspired original schematic related to Friston et al. (2024), Figs. 1 and 4; estimand: categorical dependency structure; uncertainty: none. The x-axis is the dependency or role order within each panel; the y-axis positions hidden states, observations, model factors, and optional context levels. Panel A shows a hidden 9-cell location and the corresponding categorical likelihood row $A[o, s]$, making the private sensory report explicit. Panel B shows A, B, C, and D_0 feeding the local posterior $q(s)$; Panel C shows temporal state, observation, posterior, and action order; Panel D shows optional top-down conditioned priors. The equation ribbon records the implemented state-inference form and the zero-robustness recovery identity. This deterministic formal schematic contains no fitted values, empirical sample, error band, or confidence interval.

State space: one shared latent factor I

The world holds one hidden factor: the creature's location on a square grid of side L, giving $n_s = L^2$ location states. Our sentinel world uses the 3×3 cardinality illustrated in Friston et al. (2024), Fig. 1, so $n_s = 9$ — the cardinality `pomdp.N_LOCATIONS` exposes and `experiment_config` carries as `n_locations`. This single location factor is precisely the latent the colony gossips about: it is the shared argument of the log-linear pool (eq. 6) and of every belief-sharing round (eq. 10). Fixing one hidden factor keeps the recovery limits of sec. 6 closed-form and exactly testable, rather than approximated.

Four categorical tensors: likelihood, transitions, preferences, priors I

The generative model is the tuple (A, B, C, D_0) in the discrete active-inference convention: a categorical probability mass function is a non-negative vector summing to one, and a likelihood matrix is shape (n_o, n_s) whose columns (indexed by hidden state) are categorical.

Observation likelihood $A = P(o \mid s)$. Each sentinel observes the creature's cell through a noisy sensor. With probability `acuity` the sensor reports the true cell; the residual mass $1 -$ acuity spreads uniformly over the other $n_s - 1$ cells. With outcome cardinality $n_o = n_s$, the single location modality is one (n_s, n_s) matrix:

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$$A_{os} = \begin{cases} \text{acuity}, & o = s, \\ \dfrac{1 - \text{acuity}}{n_s - 1}, & o \neq s, \end{cases} \qquad \textstyle\sum_o A_{os} = 1. \tag{1}$$

The acuity in eq. 1 tunes how peaked the sensor is: high acuity gives a near-diagonal A that pins the creature; the belief-sharing study deliberately runs the colony at the low acuity $\text{acuity} = 0.55$, where no single sentinel can resolve the location alone and the colony must pool evidence to do so. When a seeded generator is supplied, each sentinel's acuity is jittered by a small non-negative perturbation, so a colony carries slightly heterogeneous likelihoods while every column remains a proper pmf.

Transition tensor $B = P(s' \mid s, u)$. The creature moves on the grid under three control paths — `still`, `left`, `right` — so B has shape (n_s, n_s, n_u) with $n_u = 3$. `still` is the deterministic self-loop; `left` and `right` decrement and increment the grid column, saturating at the walls (a wall-adjacent move in the wall's direction is a self-loop). All three controls act on the column index alone, so the creature's row is preserved and its motion is confined to the horizontal axis of the grid — a deliberate one-dimensional control over the two-dimensional location factor. Every slice $B_{\cdot\cdot u}$ is column-normalized by construction, so the transition is a valid categorical for each action.

Log-preference C. The sentinel prefers to *see* the creature near the den (the center cell), encoded as a log-preference vector of shape $(n_o, 1)$ with a positive bump on the center outcome and zero elsewhere. The preferred-outcome distribution that the expected-free-energy decomposition of sec. 5.15 uses is $p_C(o) = \mathrm{softmax}(C)[o]$.

Initial prior D_0. The creature is believed to start at the grid center, so D_0 of shape $(n_s, 1)$ places unit mass on the center cell. D_0 enters state inference as the log-prior of the one-step variational update (eq. 2 below).

The columns-are-pmfs invariant is not assumed — it is pinned by ISC-15, which checks that every column of A and of each $B_{\cdot\cdot u}$ sums to one.

One-step variational state inference in the grid world I

Given an observation o, a sentinel forms a posterior over the creature's location by a single softmax step (Friston et al. (2024), Eq. 4): the log-prior plus the additive log-likelihood message, summed over any conditionally independent modalities m,

$$q(s) \;=\; \mathrm{softmax}\Big(\ln D_0(s) \;+\; \textstyle\sum_m \ln A_m[o_m\,,\,s]\Big). \tag{2}$$

The message $\ln A_m[o_m, \cdot]$ is the row of A_m that the observed outcome selects; summing messages over modalities makes each modality an additive evidence term — the categorical product-of-experts. The companion variational free energy, the scalar eq. 2 minimizes, is

One-step variational state inference in the grid world I

$$F[q] = \mathbb{E}_q[\ln q(s) - \ln D_0(s) - \textstyle\sum_m \ln A_m[o_m, s]] = \mathrm{KL}(q \,\|\, D_0) - \mathbb{E}_q[\textstyle\sum_m \ln A_m[o_m, s]], \quad (3)$$

reported in nats. The one-step posterior of eq. 2 is its unique minimizer, where F equals the negative log model evidence. Both live in `belief_updating.infer_states` and `belief_updating.vfe`, and the free energy of eq. 3 is the quantity the communicating-versus-incommunicado colony comparison of sec. 5.22 scores.

This inference step is not a separate mechanism bolted onto the colony: it is the $L = \mathrm{NLL}$, learning-rate-1 special case of the generalized-Bayes posterior eq. 1, reusing the same locked softmax. That client identity recovers the stated categorical Bayes substrate at its trusting limits. The separate server identity in sec. 5.8 then yields the project log-linear pool under its qualified Eq. 7 message-combination bridge; together these do not recover the complete Friston protocol (sec. 6).

The categorical POMDP loop in fig. 2 separates the common latent-state substrate from the federation transport. In the sentinel interpretation, an agent is a location-sensitive observer: the hidden state is one of 9 cells, the private outcome is a noisy categorical report of that location, and the agent sends its posterior over the location rather than the report itself. The flat belief-sharing studies use the observation, posterior, and communication subset; the moving-world extension also executes transition and EFE-guided action paths. The diagram therefore gives readers the active-inference context without turning a conceptual loop into a claim that every study estimates every latent or policy quantity.

Hidden-state to action loop: the POMDP substrate I

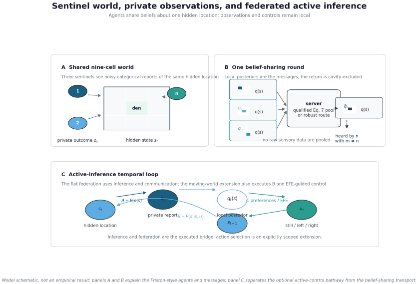

Figure 2: Sentinel-world and active-inference loop. Source relation: source-inspired original schematic related to Friston et al. (2024), Figs. 1 and 4; estimand: POMDP message-and-action sequence; uncertainty: none. The x-axis is the POMDP cycle in Panel C from hidden state through observation, posterior, action, and next state; the y-axis separates the shared-world, belief-sharing, and temporal-loop panels. Panel A shows three agents viewing the same 9-cell hidden world through private, noisy categorical observations; raw observations remain local. Panel B shows those local posteriors entering a log-linear-pool or robust server and returning as a cavity-excluded consensus. Panel C gives the POMDP cycle from hidden state s_t through observation o_t, posterior $q_t(s)$, action u_t, and transition to s_{t+1}, with A, B, and C marking the likelihood, transition, and preference factors. The flat studies execute the inference-sharing branch; the moving-world extension also executes transitions and EFE-guided actions. This is a deterministic model schematic, not an uncertainty-bearing empirical result.